



Emergency Department Patient Flow - Model Documentation

Prepared for the module: Simulation of Production Systems (SS26)

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Model file: EdPatientFlow.alp

Platform: AnyLogic (Process Modeling Library, Discrete-Event Simulation)

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1. Context & Purpose

- **Problem statement:** How should Emergency Department (ED) staffing and bed capacity (triage nurses, doctors, ED beds, fast-track rooms, inpatient beds) be sized so that patient length of stay (LOS) and resource utilization stay at acceptable levels under realistic patient arrival and acuity patterns?
- **Stakeholders / users:** ED operations managers, hospital capacity planners, nursing and physician leadership evaluating staffing/bed decisions.
- **Intended decisions:** Right-sizing of ED bed capacity, inpatient bed capacity, doctor headcount, triage nurse headcount, and fast-track room capacity; and whether to operate a fast-track pathway for low-acuity patients.

2. System Definition

- **System boundary:** In scope: patient arrival, triage, acuity-based routing (fast track vs. main ED), physician exam, optional diagnostics, results review, admit/discharge decision, inpatient ward stay, and departure. Out of scope: ambulance transport, other hospital departments, physical layout/movement, and detailed clinical care pathways.
- **Level of abstraction:** Aggregate discrete-event process; each step is represented by a statistical service-time distribution rather than a detailed clinical or spatial model. No pedestrian/GIS layer is used — resources are logical pools, not physical rooms with travel time.
- **Time horizon & time unit:** 7 days (10,080 minutes) per simulation run; base time unit is minutes.
- **Spatial scope:** None — the model is a non-spatial flowchart (Process Modeling Library) representing a single generic ED department.

3. Objectives & KPIs

- **Primary objectives:** (1) Minimize patient length of stay across pathways; (2) keep key resources (ED beds, doctors, inpatient beds, triage nurses, fast-track rooms) adequately, but not excessively, utilized; (3) evaluate the effect of capacity changes (especially inpatient and ED bed capacity) on flow.

KPIs / performance measures: **KPIs / performance measures:**

KPI	Definition	Unit / Collector
LOS – Overall	Time from arrival to any exit (discharge or ward transfer)	minutes — losOverall
LOS – Fast track	Time from arrival to exit via fast-track pathway	minutes — losFastTrack
LOS – Admitted	Time from arrival to exit after inpatient ward stay	minutes — losAdmitted
Resource utilization	Busy time share of ED beds, doctors, fast-track rooms, inpatient beds, triage nurses	statEDBed, statDoctor, statFastTrack, statInpatient, statTriage

4. Modeling Approach (AnyLogic-specific)

- **Paradigm(s) used:** Discrete Event (Process Modeling Library only — no Agent Based or System Dynamics elements).
- **Main AnyLogic libraries used:** Process Modeling Library — Source, Service, SelectOutput, Seize, Release, Delay, Sink, ResourcePool blocks.
- **Key design choice:** ED patient flow is naturally a sequence of discrete process steps competing for a small number of shared, countable resources (beds, doctors, nurses, rooms). Discrete-event process modeling captures queueing and resource contention directly, without needing the spatial/movement detail that agent-based or GUI-based pedestrian modeling would add.

5. Model Structure

- **Entities / agents:** Patient agent, with parameters: arrivalTime (set on creation), acuity (1 = most severe, 2 = moderate, 3 = least severe), needsDiagnostics (boolean).

Resources (ResourcePools):**Resources (ResourcePools):**

Resource pool	Capacity parameter	Default
EDBedPool	edBedCapacity	15
DoctorPool	doctorCapacity	3
FastTrackRoomPool	fastTrackCapacity	5
InpatientBedPool	inpatientCapacity	23
TriageNurse	triagenurseCapacity	2

Process flow (high-level):**Process flow (high-level):**

- source → Triage (Service, seizes TriageNurse, delay ~ triangular(2, 4, 6) min)
- Triage → RoutebyAcuity (SelectOutput): condition = acuity == 3 AND useFastTrack
 - If true → FastTrackRoom (Service, seizes FastTrackRoomPool, queue capacity 3, delay ~ triangular(10, 20, 30) min) → sink
 - If false → EDBed (Seize EDBedPool) → PhysicianExam (Service, seizes DoctorPool; delay depends on acuity: acuity 1 ~ triangular(20,30,45), acuity 2 ~ triangular(15,25,35), acuity 3 ~ triangular(10,...)) min
 - PhysicianExam → diagnosisChecking (SelectOutput): condition = needsDiagnostics
 - If true → delay1 (diagnostic wait, ~ triangular(20, 40, 60) min) → ResultReview (Service, seizes DoctorPool, ~ triangular(5, 10, 15) min) → AdmitDischarge
 - If false → straight to AdmitDischarge
 - AdmitDischarge (SelectOutput): condition = uniform(0,1) < admission probability by acuity (acuity 1 = 0.50, acuity 2 = 0.20, acuity 3 = 0.05)
 - If admitted → InpatientBed (Seize InpatientBedPool) → release (frees EDBed) → WardStay (Delay ~ exponential, mean 2,880 min ≈ 2 days, capacity 20) → release1 (frees InpatientBed) → sink1
 - If not admitted → discharge (frees EDBed) → sink2
- **State logic:** On arrival, acuity is drawn once (10% probability acuity = 1, 30% acuity = 2, 60% acuity = 3) and needsDiagnostics is set true whenever acuity ≤ 2. These two attributes drive every downstream routing decision (fast track, diagnostics, admission).

6. Inputs & Data

Input parameters (controllable, with defaults):**Input parameters (controllable, with defaults):**

Parameter	Type	Default
useFastTrack	boolean	true
meanInterarrivalTime (min)	double	10
edBedCapacity	int	15
doctorCapacity	int	3
fastTrackCapacity	int	5

inpatientCapacity	int	23
triagenurseCapacity	int	2

Distributions (stochastic elements):**Distributions (stochastic elements):**

- Inter-arrival time: exponential(1 / meanInterarrivalTime) minutes — constant-rate Poisson arrivals (no time-of-day variation, though an unused rateSchedule parameter exists on the Source block)
- Acuity assignment: uniform(0,1) draw → 10% acuity 1, 30% acuity 2, 60% acuity 3
- Triage duration: triangular(2, 4, 6) min
- Fast-track service: triangular(10, 20, 30) min
- Physician exam: triangular by acuity — 1: (20,30,45); 2: (15,25,35); 3: (10,...)
- Diagnostic wait: triangular(20, 40, 60) min
- Result review: triangular(5, 10, 15) min
- Admission decision: uniform(0,1) vs. acuity-based probability (0.50 / 0.20 / 0.05)
- Ward stay: exponential with mean 2,880 min (~2 days)
- **Data sources:** All distribution shapes and parameter values are modeler assumptions / illustrative benchmarks consistent with typical ED literature; none are shown as calibrated against a specific hospital's historical data.
- **Calibration choices:** None applied — this is flagged as a limitation below.

7. Assumptions & Limitations

Assumptions (explicit list):**Assumptions (explicit list):**

- No balking or reneging — patients never leave a queue voluntarily.
- Constant staff capacity — no shifts, breaks, or time-varying schedules for doctors, nurses, or beds.
- No transport or layout delay — resources are logical pools; there is no travel time between triage, exam, and ward.
- Diagnostics are represented as a single generic wait (Delay block), not a distinct constrained lab/imaging resource.
- Ward stay is modeled as one aggregate exponential delay, not broken down by diagnosis or clinical pathway.
- Arrival rate is constant over the week — no day-of-week or hour-of-day demand pattern.
- **Limitations / validity range:** Results are useful for exploring relative, directional trade-offs between capacity settings (e.g., inpatient beds vs. LOS/utilization) under simplified assumptions. They should not be used as absolute predictions of a real ED's performance until arrival patterns, service-time distributions, and admission probabilities are validated against real hospital data.

8. Experiments & Scenarios

- **Baseline scenario:** "Simulation" experiment — fixed seed (1), single run, 10,080-minute (7-day) horizon, all parameters at their default values listed in Section 6.

Alternative scenarios (Parameter Variation experiments) — five in total, one per resource pool:**Alternative scenarios (Parameter Variation experiments) — five in total, one per resource pool:**

Experiment	Varied parameter	Range
ParametersVariation_InPatientCapacity	inpatientCapacity	10 → 40, step 1
ParameterVariation_EDBed	edBedCapacity	15 → 30, step 1
ParametersVariation_DoctorCapacity	doctorCapacity	2 → 8, step 1

ParametersVariation_NurseCapacity	triagenurseCapacity	1 → 5, step 1
ParametersVariation_FastTrack	fastTrackCapacity	2 → 10, step 1

- **Experiment type:** Parameter Variation (range sweep) for all five alternative scenarios; the baseline uses a single Simulation experiment run.
- **Replication plan:** No explicit warm-up period is configured; each replication runs the full 7-day horizon from a fresh start with fixed seed = 1. Replication counts ("NumberOfRuns" / "ReplicationPerIteration") are configured as shown above — the ReplicationsProperties block also lists UseReplication = false, which should be reconciled before treating the reported means as statistically robust confidence intervals.

9. Outputs & Analysis

- **Output metrics:** Base run: losOverall, losFastTrack, losAdmitted (minutes); statEDBed, statDoctor, statFastTrack, statInpatient, statTriage (resource utilization). Each of the five sweep experiments pairs its capacity parameter with a matching utilization dataset and LOS dataset:

Sweep experiment	Utilization dataset	LOS dataset (vs. capacity)
ParametersVariation_InPatientCapacity	utilInpatient	losAdmittedByCapacity
ParameterVariation_EDBed	utilEDBed	losOverallByCapacity
ParametersVariation_DoctorCapacity	utilDoctor	losOverallByCapacity
ParametersVariation_NurseCapacity	utilTriage	losOverallByTriage
ParametersVariation_FastTrack	utilFastTrack	losFastTrackByCapacity

- **How results are aggregated:** Each experiment's AfterSimulationRunCode records the mean utilization and mean LOS for that run (e.g., root.statInpatient.mean(), root.losAdmitted.mean()) as one point per capacity level, plotted as a capacity-sweep curve. It also writes the same two numbers to the console via traceln(). No percentile or time-weighted-vs-observation-based split is currently distinguished in the outputs.
- **Decision interpretation:** For each resource pool, look for the capacity level where mean utilization falls to a comfortable operating band (e.g., below ~85%) while the paired LOS curve flattens out — this marks the point of diminishing returns from adding more of that resource, and is a reasonable candidate for a capacity/staffing recommendation. Running all five sweeps together also shows which resource (ED beds, doctors, inpatient beds, triage nurses, or fast-track rooms) is the binding constraint on overall LOS.

Baseline Results ("Simulation" experiment, seed = 1, 7-day horizon):

Output	Mean	Samples
LOS – Overall	191.64	919
LOS – Fast Track	24.16	603
LOS – Admitted	1,779.35	76
EDBed utilization (statEDBed)	0.410	1,009
Doctor utilization (statDoctor)	0.424	1,009
FastTrack utilization (statFastTrack)	0.240	1,009
Inpatient utilization (statInpatient)	0.750	1,009
Triage utilization (statTriage)	0.206	1,009

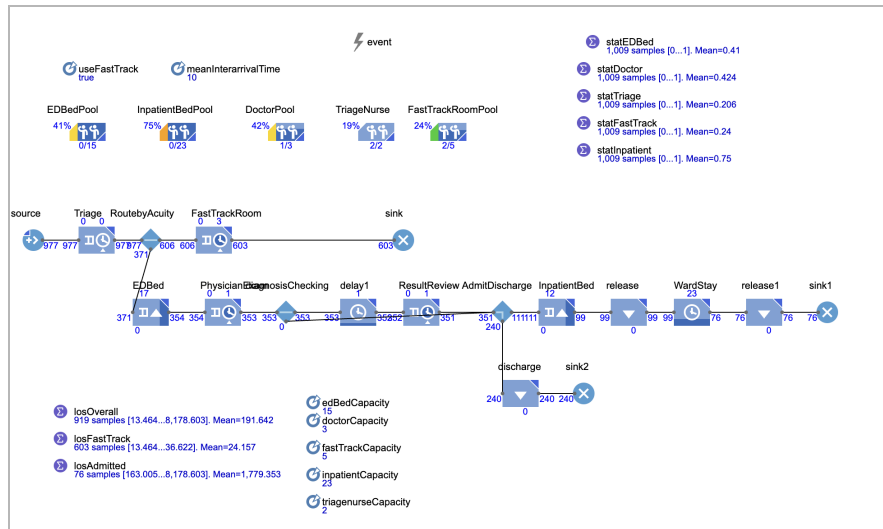


Figure: baseline simulation dashboard showing resource pool occupancy, utilization statistics, and LOS distributions for the default parameter set.

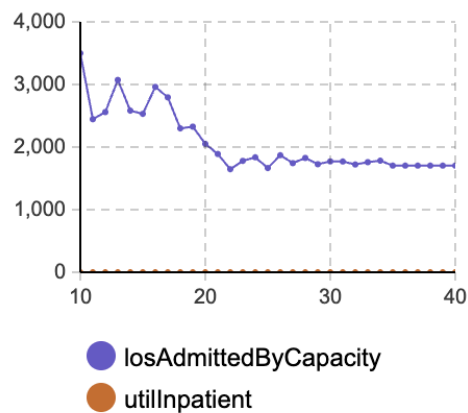
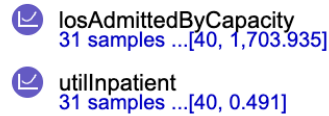
Parameter Variation Sweep Results:

The following tables and charts report the mean utilization and mean LOS recorded by each Parameter Variation experiment at every tested capacity level (order matches Section 8).

Inpatient Capacity Sweep (ParametersVariation_InPatientCapacity)

inpatientCapacity	Utilization (mean)	LOS – Admitted (mean, min)
10	0.917	3,499.1
11	0.910	2,445.1
12	0.902	2,559.4
13	0.895	3,072.2
14	0.889	2,582.2
15	0.883	2,530.8
16	0.877	2,961.4
17	0.872	2,794.4
18	0.866	2,298.6
19	0.852	2,326.4
20	0.807	2,052.0
21	0.790	1,889.4
22	0.764	1,646.5
23	0.750	1,779.4
24	0.733	1,837.1
25	0.692	1,664.6
26	0.700	1,870.0
27	0.685	1,743.3
28	0.672	1,826.2
29	0.652	1,726.3
30	0.626	1,771.3
31	0.611	1,769.8
32	0.605	1,721.8
33	0.578	1,759.1

inpatientCapacity	Utilization (mean)	LOS – Admitted (mean, min)
34	0.595	1,782.5
35	0.562	1,703.9
36	0.546	1,703.9
37	0.531	1,703.9
38	0.517	1,703.9
39	0.504	1,703.9
40	0.491	1,703.9

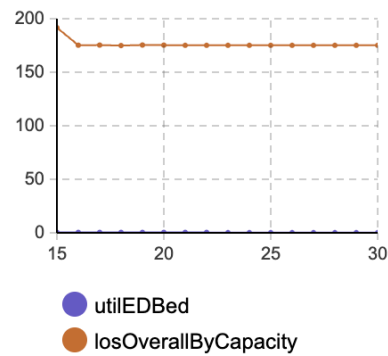


Utilization declines steadily as inpatient capacity increases; mean LOS – Admitted drops sharply up to ~capacity 22–24 and then plateaus around 1,700–1,800 minutes, with LOS flattening beyond capacity 35.

ED Bed Capacity Sweep (ParameterVariation_EDBed)

edBedCapacity	Utilization (mean)	LOS – Overall (mean, min)
15	0.410	191.6
16	0.341	175.2
17	0.322	175.3
18	0.307	174.9
19	0.290	175.4
20	0.276	175.3
21	0.262	175.1
22	0.250	175.1
23	0.240	175.1

edBedCapacity	Utilization (mean)	LOS – Overall (mean, min)
24	0.230	175.1
25	0.220	175.1
26	0.212	175.1
27	0.204	175.1
28	0.197	175.1
29	0.190	175.1
30	0.184	175.1



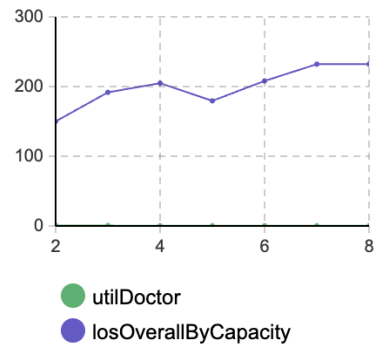
Overall LOS drops from the default (capacity 15) to capacity 16 and then flattens near ~175 minutes for all higher capacities — ED beds are not the binding constraint on overall LOS beyond the default setting.

Doctor Capacity Sweep (ParametersVariation_DoctorCapacity)

doctorCapacity	Utilization (mean)	LOS – Overall (mean, min)
2	0.713	149.9
3	0.424	191.6
4	0.303	204.8
5	0.261	179.5
6	0.227	208.0
7	0.199	232.1
8	0.174	232.1

utilDoctor
7 samples ...[8, 0.174]

losOverallByCapacity
7 samples ...[8, 232.117]



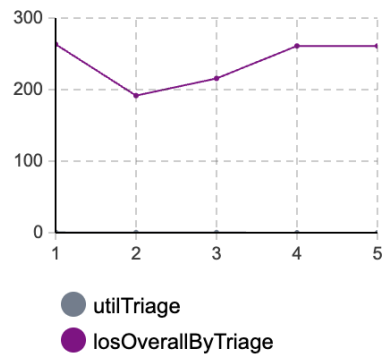
Counter-intuitively, mean overall LOS rises as doctor capacity increases beyond 2–3 — consistent with a downstream bottleneck (e.g., inpatient beds) capping throughput regardless of how quickly patients are examined.

Triage Nurse Capacity Sweep (ParametersVariation_NurseCapacity)

triagenurseCapacity	Utilization (mean)	LOS – Overall (mean, min)
1	0.379	263.4
2	0.206	191.6
3	0.139	215.6
4	0.105	261.0
5	0.084	261.0

utilTriage
5 samples ...[5, 0.084]

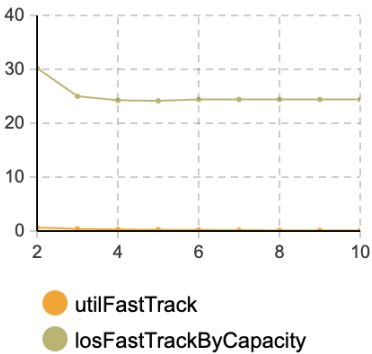
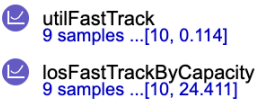
losOverallByTriage
5 samples ...[5, 261.044]



Only 5 data points are available. LOS is lowest at 2 nurses (the default) and highest at 1 or 4–5 — the non-monotonic pattern likely reflects the small sample and single-seed replication rather than a genuine effect, and should not be over-interpreted.

Fast-Track Room Capacity Sweep (ParametersVariation_FastTrack)

fastTrackCapacity	Utilization (mean)	LOS – Fast Track (mean, min)
2	0.621	30.23
3	0.402	25.00
4	0.291	24.25
5	0.240	24.16
6	0.190	24.41
7	0.163	24.41
8	0.142	24.41
9	0.126	24.41
10	0.114	24.41



Fast-track LOS drops from ~30 minutes at capacity 2 to ~24 minutes by capacity 4–5 and is flat thereafter — capacity beyond ~5 rooms yields negligible further benefit for fast-track patients.

10. Verification & Validation

Verification checks (model works as intended):**Verification checks (model works as intended):**

- No explicit automated sanity checks (e.g., in/out patient-count balance) are currently coded — recommended as a next step.
- Recommended: run extreme-case tests (e.g., capacity = 1 for a resource) to confirm queues build up and drain as expected, and step through the flowchart with animation to confirm routing logic (RoutebyAcuity, diagnosisChecking, AdmitDischarge conditions) fires as intended.

Validation approach (model matches reality or expectations):**Validation approach (model matches reality or expectations):**

- No comparison to historical ED data is currently built into the model.

- Recommended: face validation with ED clinical/operations staff on distribution shapes and admission probabilities, plus the sensitivity analysis already enabled by the two Parameter Variation experiments.

11. Reproducibility

- **How to run it:** Open EdPatientFlow.alp in AnyLogic, then run the "Simulation" experiment for a single 7-day baseline run (seed = 1). Run any of the five Parameter Variation experiments — ParametersVariation_InPatientCapacity, ParameterVariation_EDBed, ParametersVariation_DoctorCapacity, ParametersVariation_NurseCapacity, or ParametersVariation_FastTrack — for the corresponding single-resource capacity-sweep analysis.
- **Key parameters to change:** Available on the Main agent's parameter panel — useFastTrack, meanInterarrivalTime, edBedCapacity, doctorCapacity, fastTrackCapacity, inpatientCapacity, triagenurseCapacity.

Known issues / TODOs:**Known issues** / **TODOs**:

- ReplicationsProperties for the inpatient-capacity sweep show UseReplication = false
- **Repository / file naming:** Model file: EdPatientFlow.alp. No external data files are referenced — all inputs are hard-coded parameters/distributions within the model.

12. Learning Reflection

- **Biggest modeling risk:** Service-time distributions and admission probabilities are illustrative assumptions rather than data-calibrated values, so capacity recommendations drawn from this model could be directionally right but numerically off versus a real ED.
- **Biggest assumption impact:** Treating arrivals as constant-rate (no time-of-day/day-of-week pattern) likely understates peak-hour crowding and resource strain, since real EDs typically see pronounced daily demand cycles.
- **What to improve with more time/data:** Calibrate arrival and service-time distributions from real ED logs; add a time-varying arrival schedule; add explicit verification checks and a warm-up period; and run multiple random seeds per scenario to report confidence intervals rather than single-seed means.